Ship Knowledge and Equipment

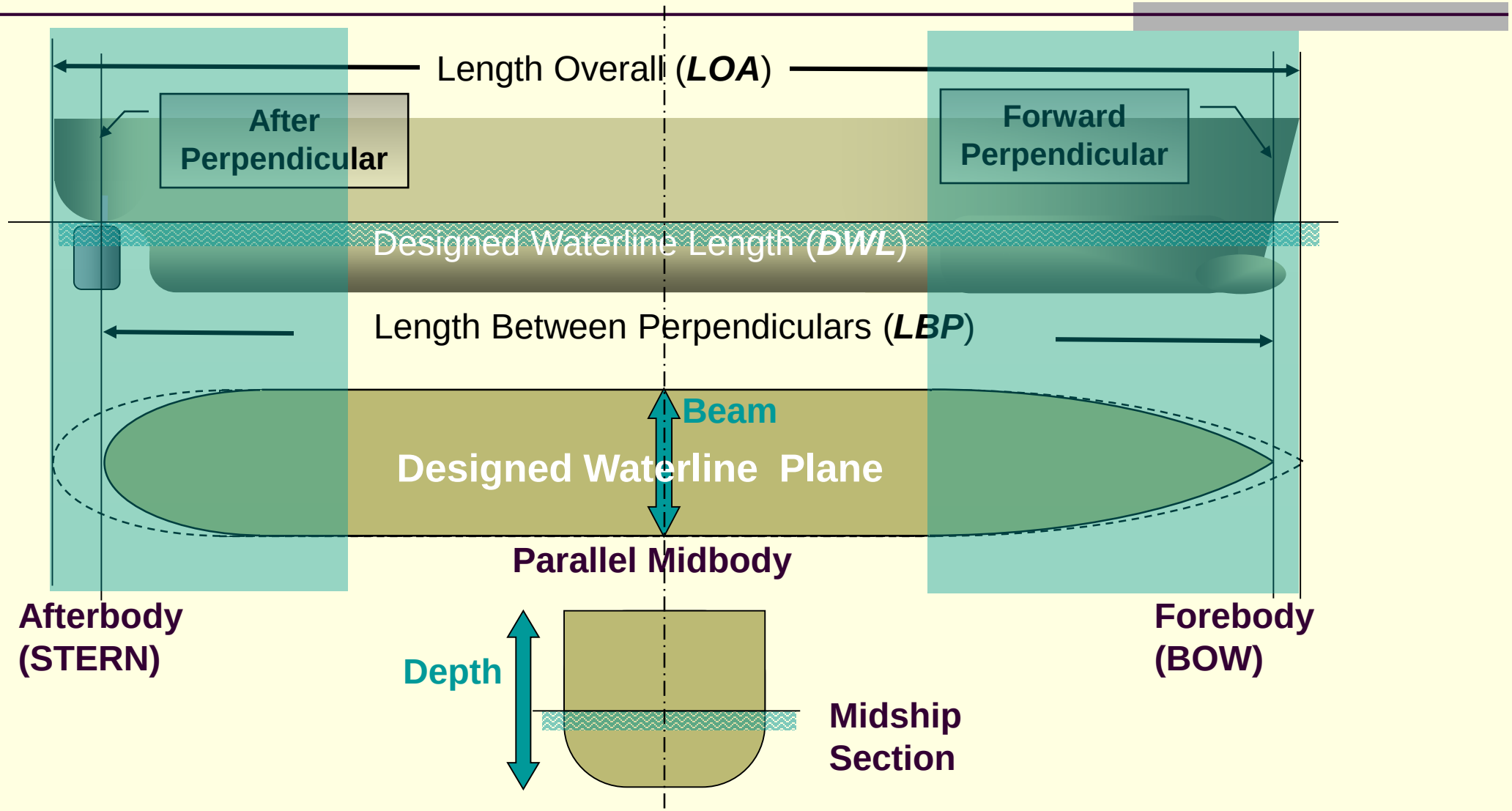
NTA LEVEL 4 & BRIDGE COURSE

SRT 04104

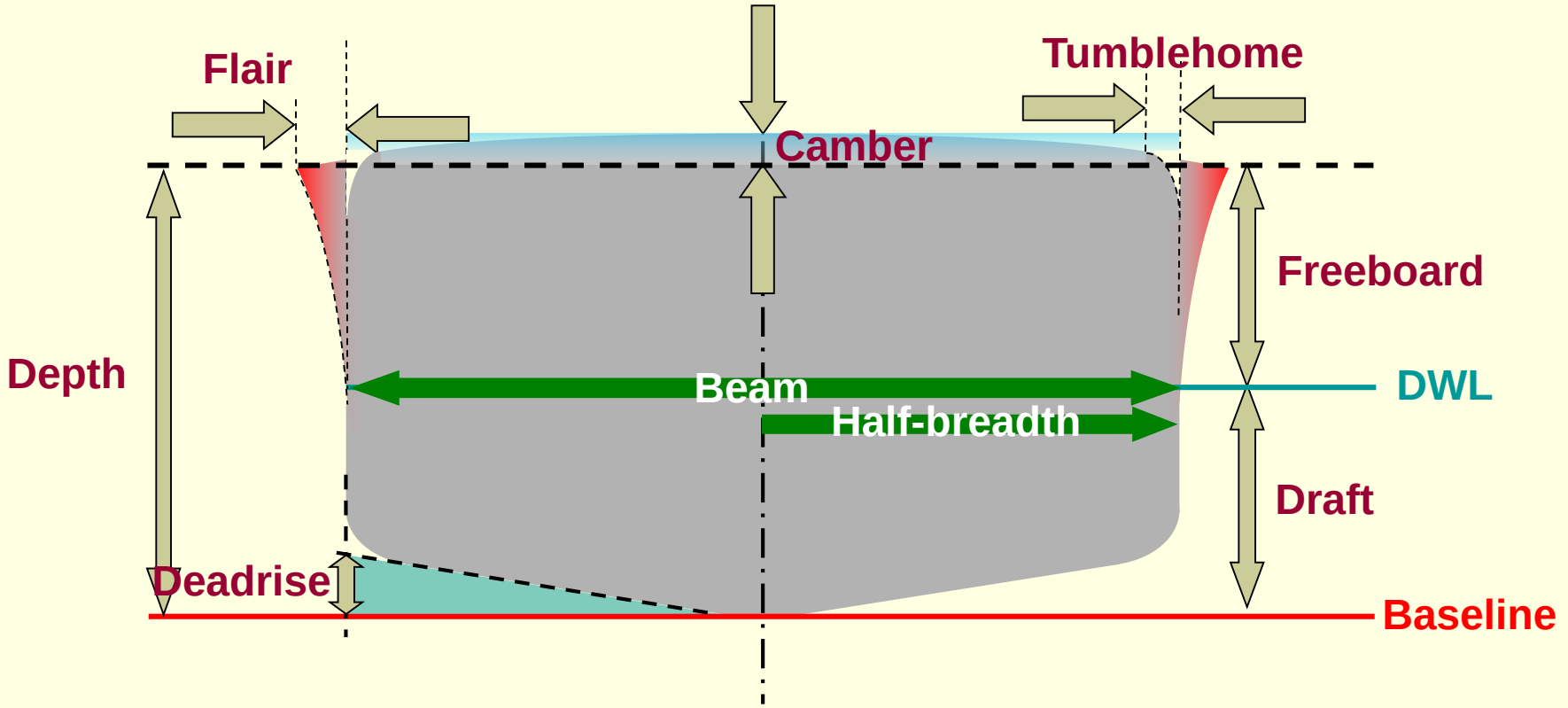
Lecture contents

- Ship Dimensions
- Section Nomenclature
- Coefficients of Form
- Types of lines
- Lines Plan

Ship Dimensions



Section Nomenclature



Form coefficients

- Form coefficients are numerical ratios which compare the ship underwater form and regular shapes having the same dimensions as the ship.
- They are used in ship design to predict ship resistance to forward motion, to estimate the ship required power to deliver the service speed.

Waterplane area coefficient C_w

- Is the ratio of the waterplane area (WPA) to the area of a rectangle having the same length (L) and breadth (B) as that of the ship at the waterline.

$$C_w = \frac{WPA}{L \times B}$$

- Note: since the ship's waterplane area is smaller compared to that of the rectangle, Cw < 1

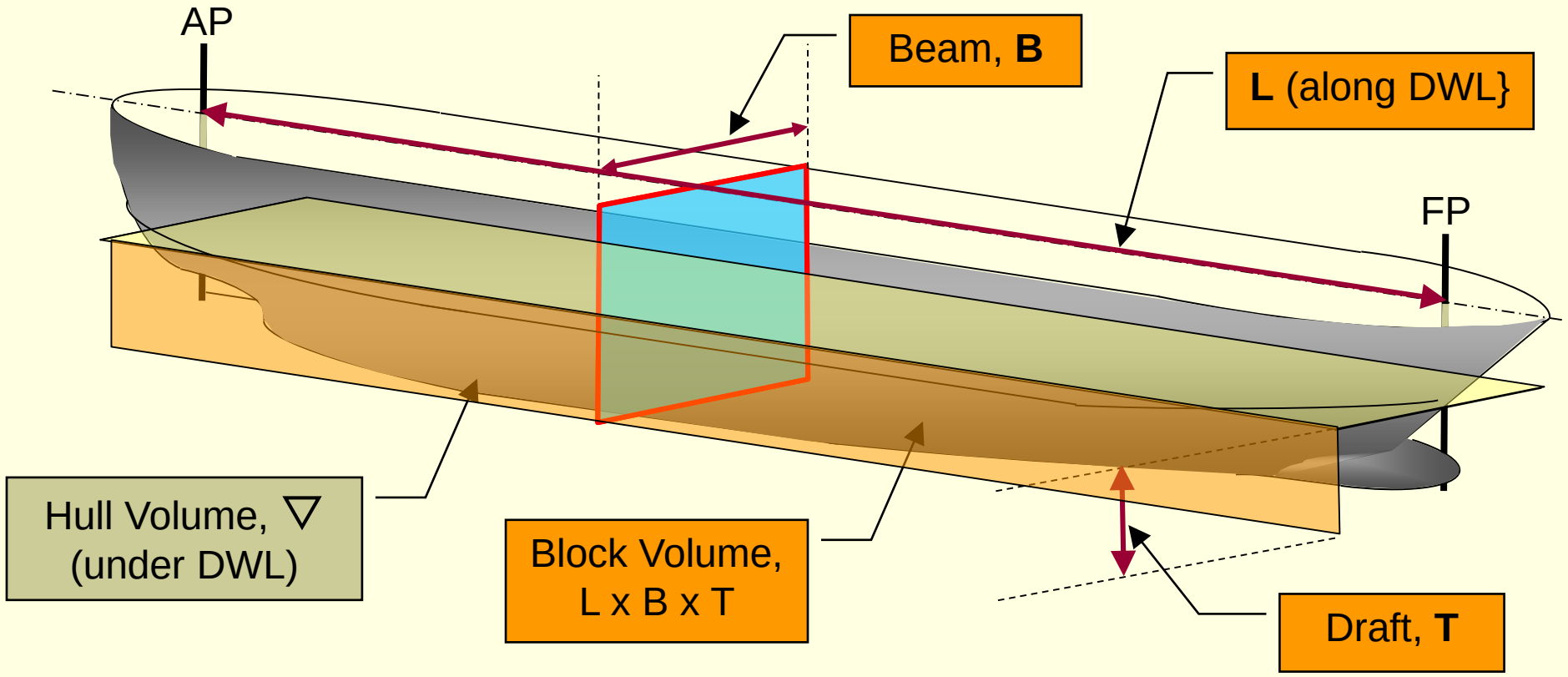
Block coefficient C_b

- Is the ratio of the underwater volume of a ship to the volume of the circumscribing block.

$$C_b = \frac{Volume\ of\ Displacement}{L \times B \times d}$$

Block Coefficient C_b

- Block Coefficient: $\mathbf{C_B} = \nabla \, / \, (L \times B \times T)$



Midship Area coefficient C_M

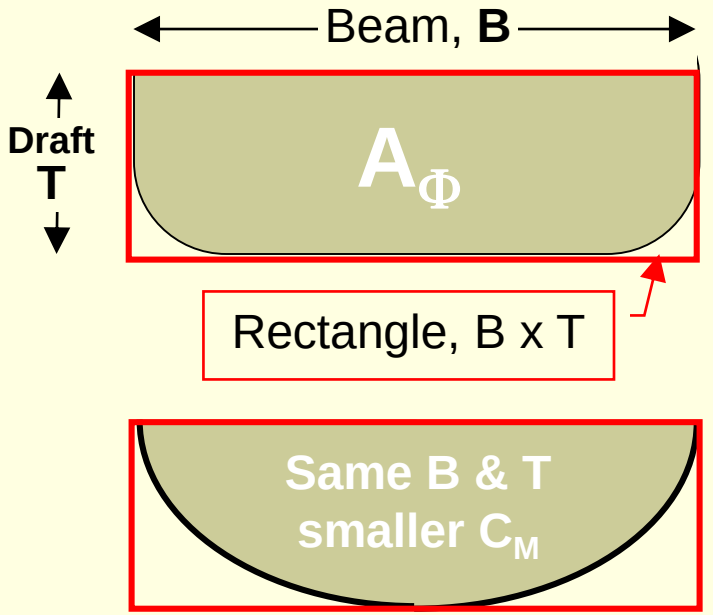
- Is the ratio of the underwater transverse area of midship section (A_m) to the product of breadth and draught.

$$\boldsymbol{C_m = \frac{A_m}{B \times d}}$$

Midship Section coefficient

- Midship Section Coefficient: $C_M = A_\Phi / (B \times T)$

- Note that $C_M = C_B / C_P$
- **COF's** are used in the early stages of design to compare characteristics of new design with existing types for estimating performance



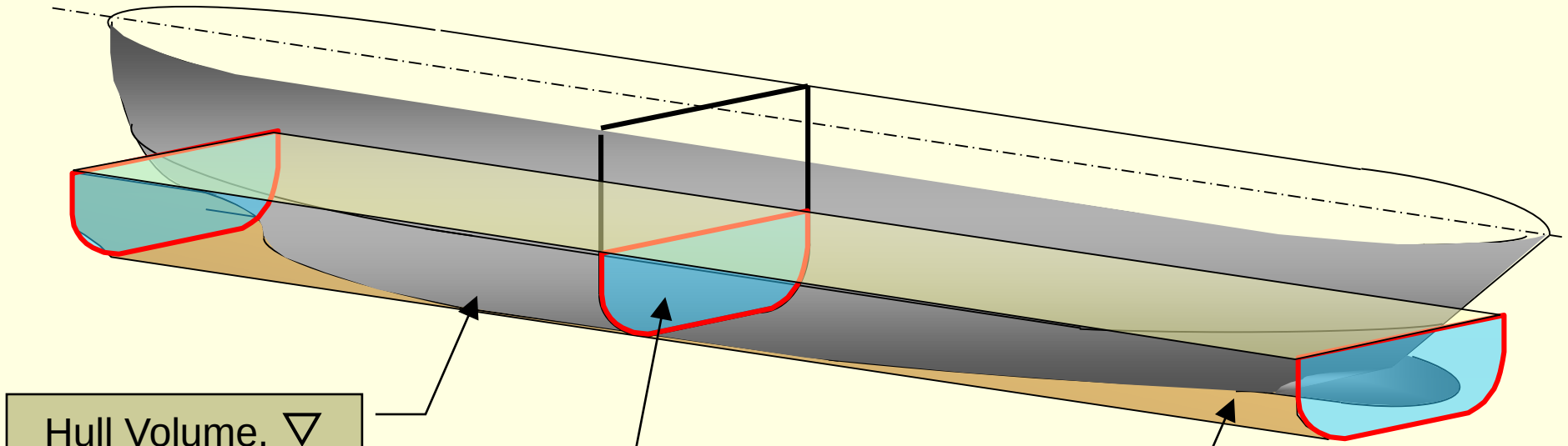
Prismatic Coefficient C_p

- Is the ratio of the underwater volume of the ship to the volume of a prism formed by the product of the underwater transverse area of the midship section and the waterline length.

$$C_p = \frac{Volume\ of\ displacement}{L \times A_m}$$

Prismatic Coefficient C_p

- Prismatic Coefficient: $\mathbf{C_P} = \nabla \, / \, (\mathbf{A_\Phi} \times L)$



Coefficient of Form

- **Note:**

$$C_m \times C_p = \frac{A_m}{B \times d} \times \frac{\text{Volume of ship}}{L \times A_m}$$

$$= \frac{\text{Volume of ship}}{L \times B \times d}$$

$$= C_b$$

$$\therefore C_m \times C_p = C_b \quad \text{or}$$

$$C_p = \frac{C_b}{C_m}$$

Tonne Per Centimeter Immersion (TPC)

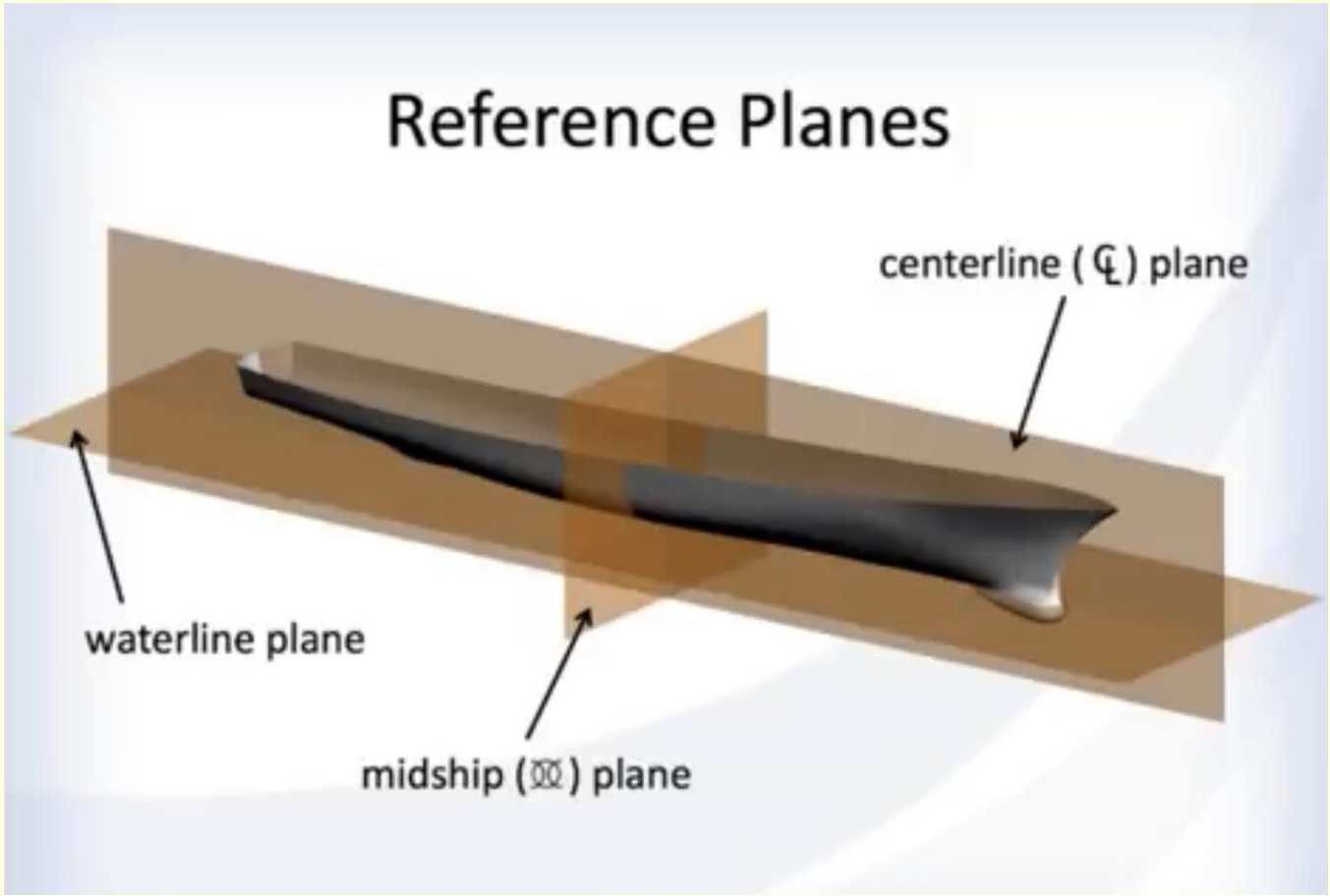
- Is the weight that must be loaded or discharged to change the ship's mean draught by one centimeter.

$$TPC = WPA(m^2) \times \frac{1}{100} \times \delta$$

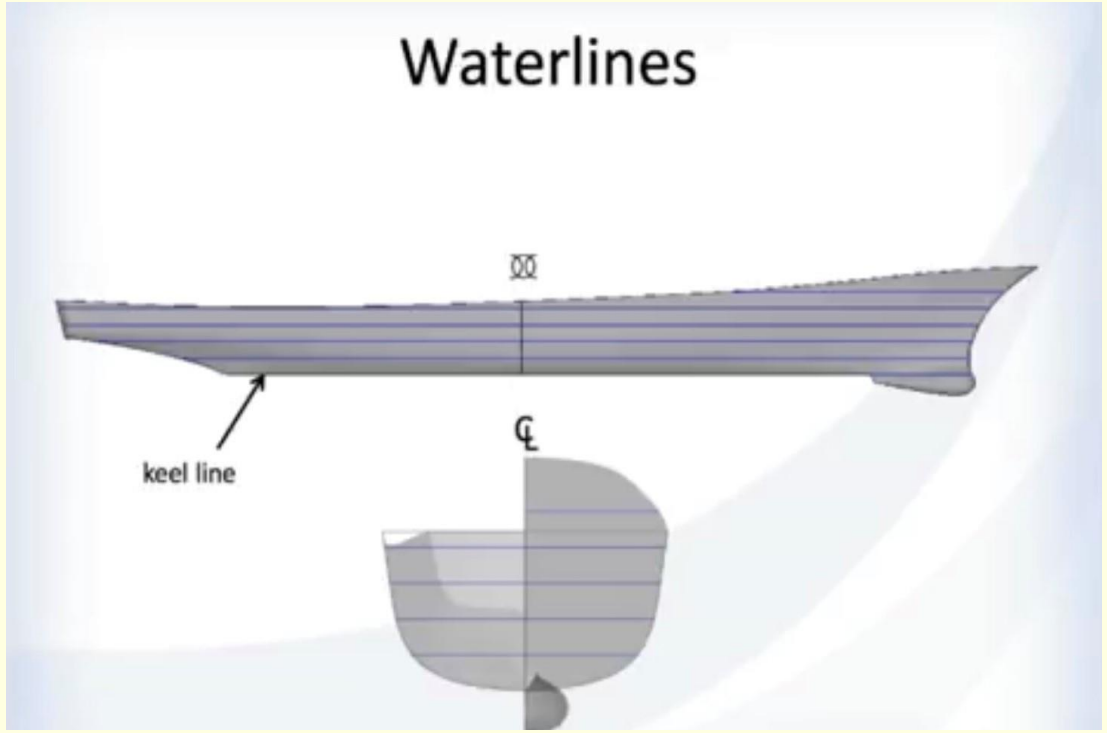
Types of Lines

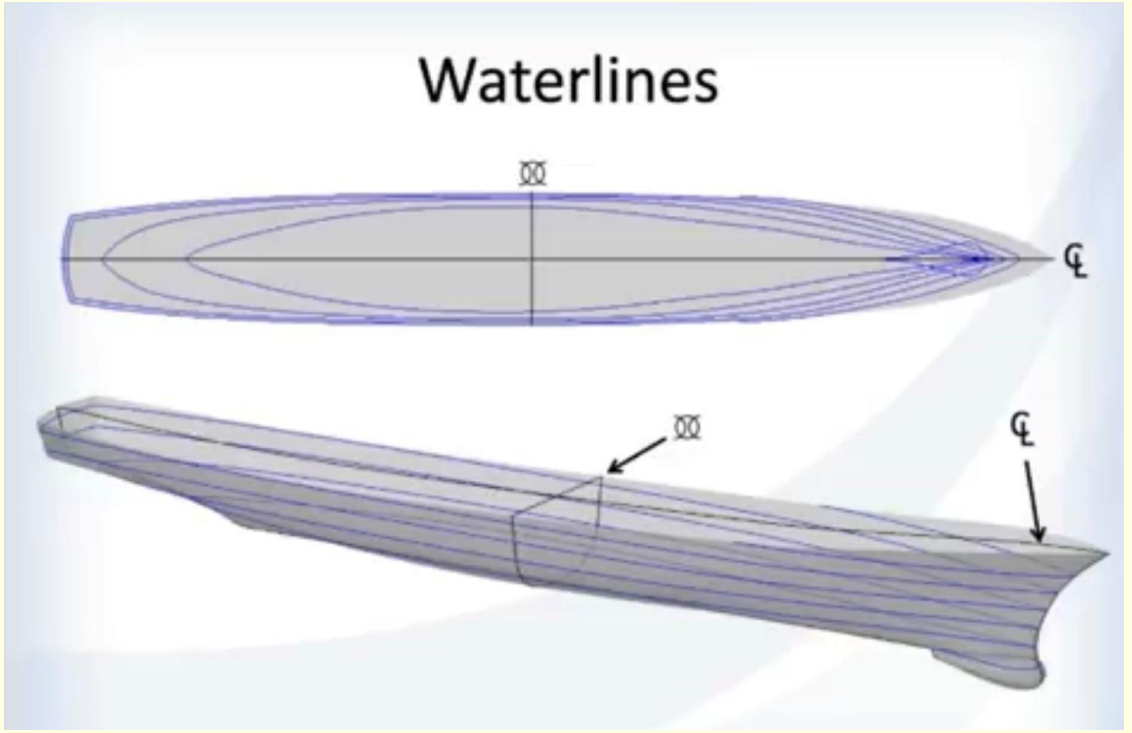
- Waterlines
- Stations or Sections
- Buttocks

Reference Planes

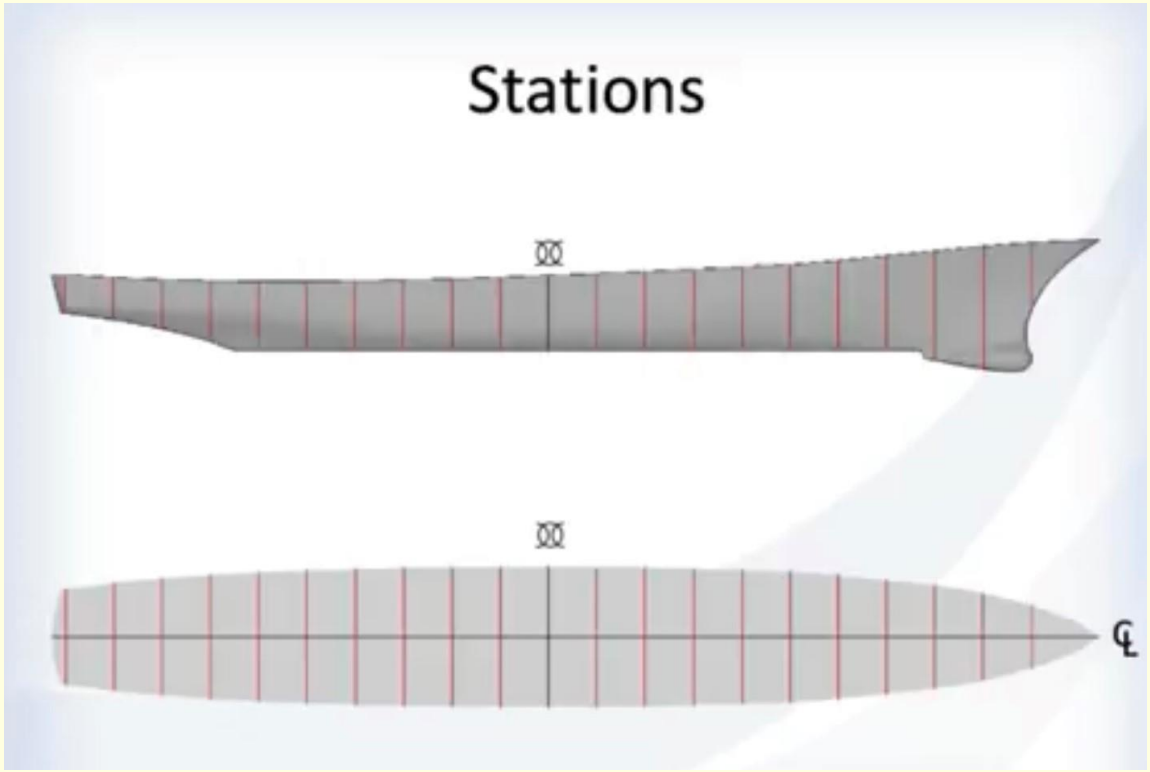


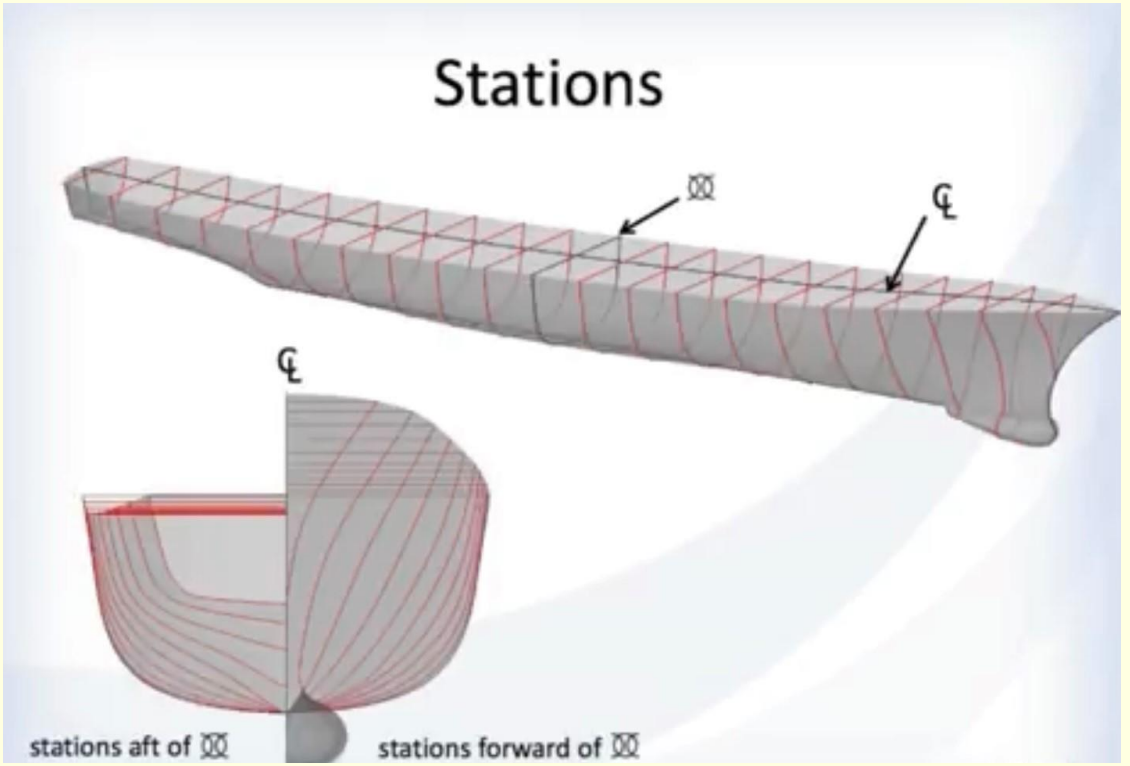
Waterlines



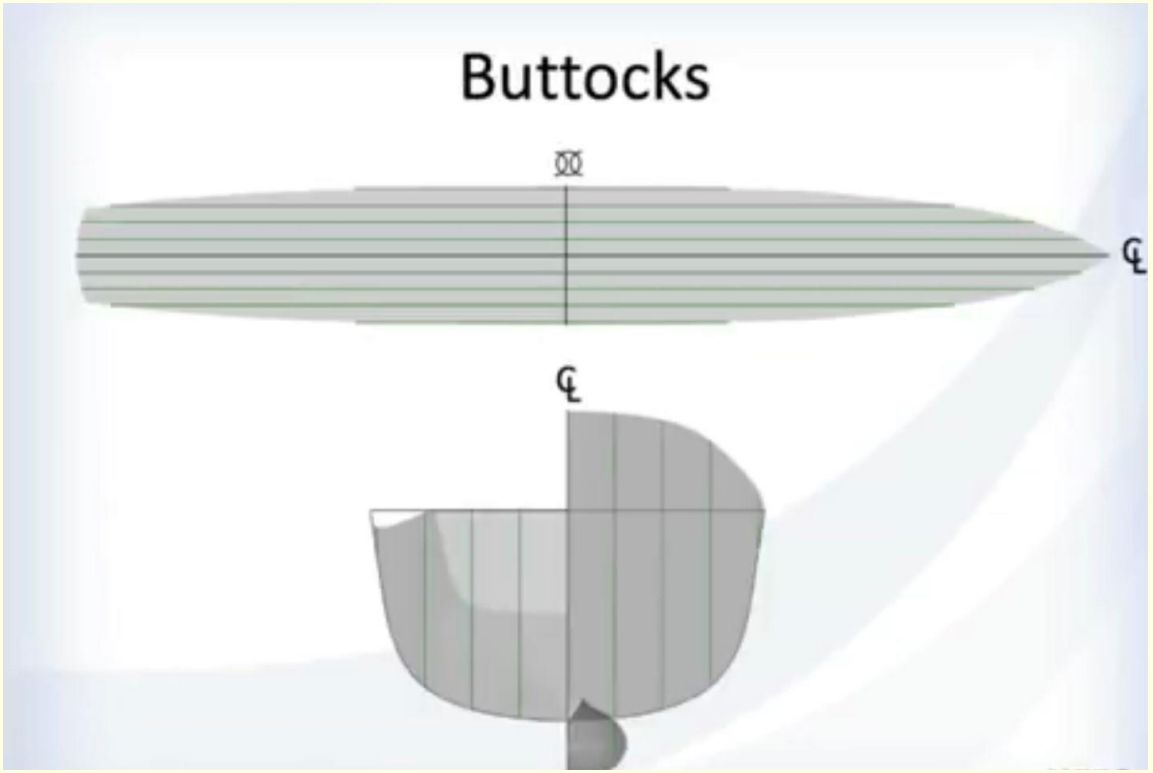


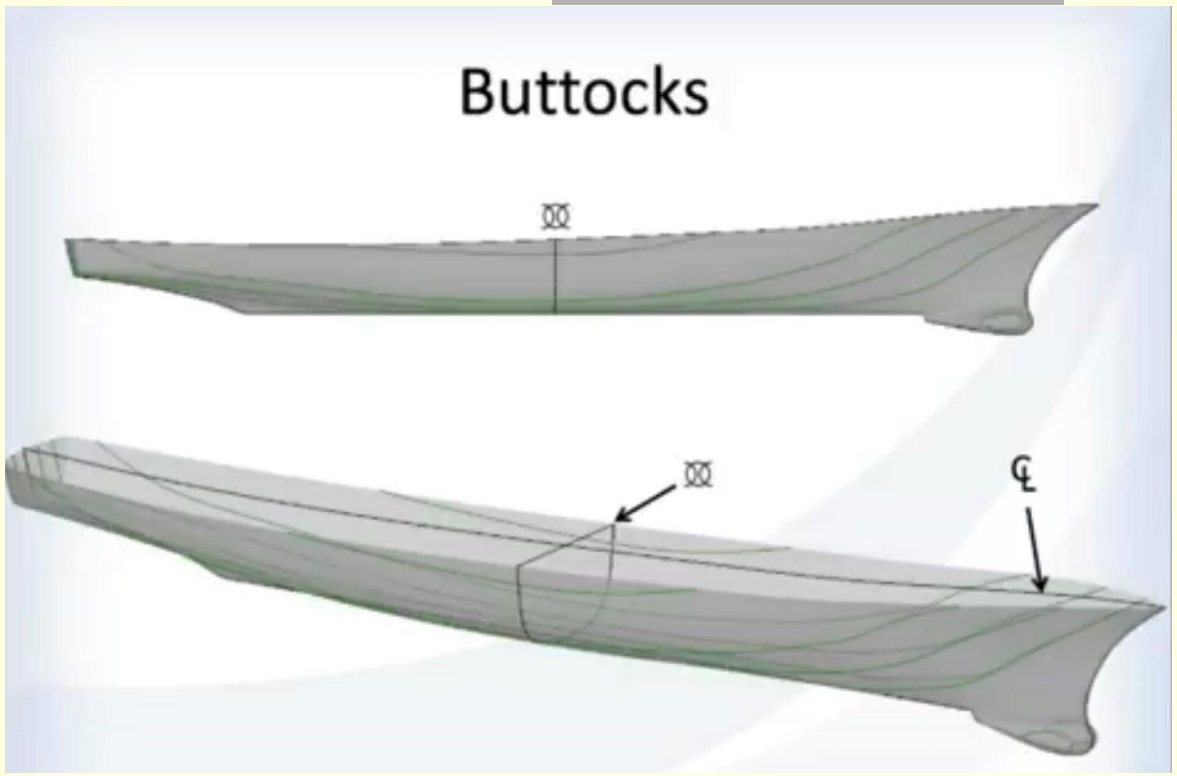
Stations





Buttocks

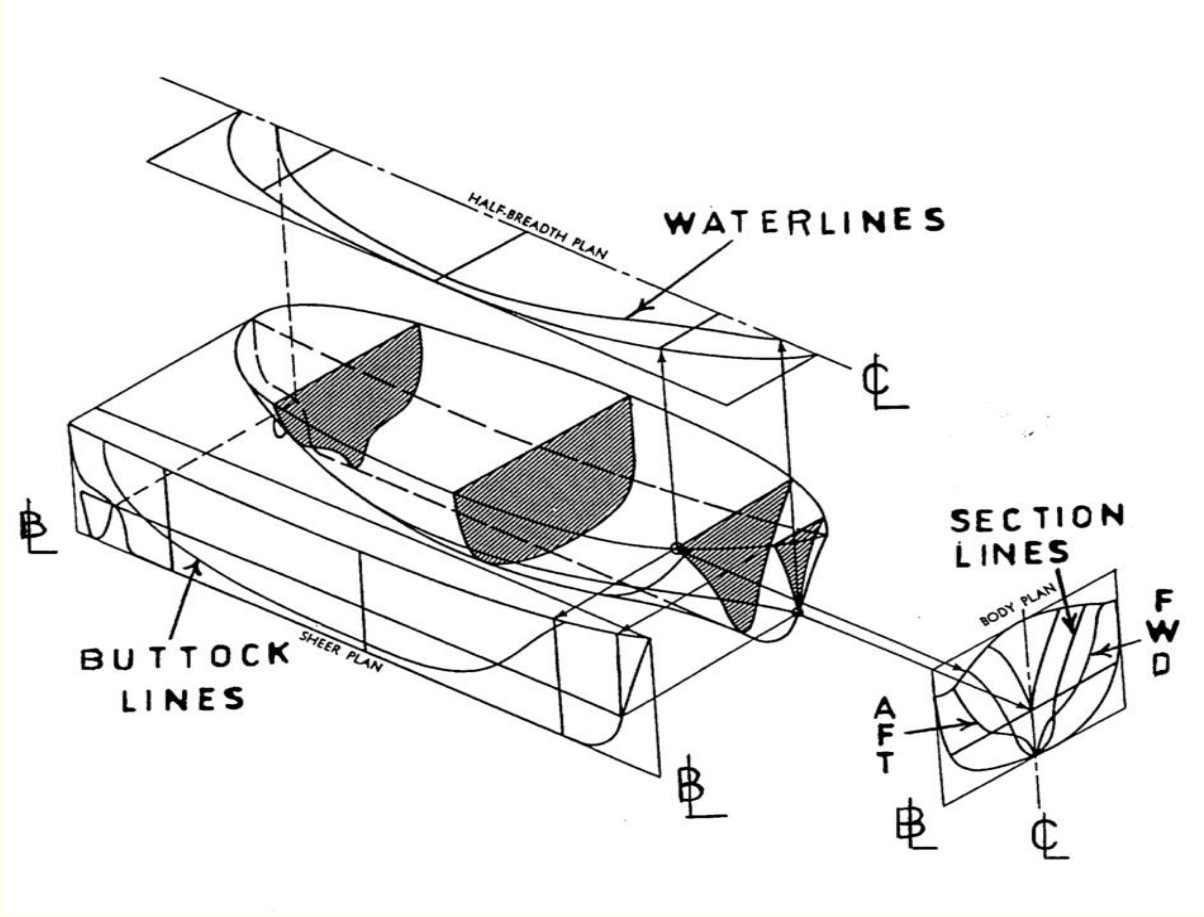




Lines drawings/plan

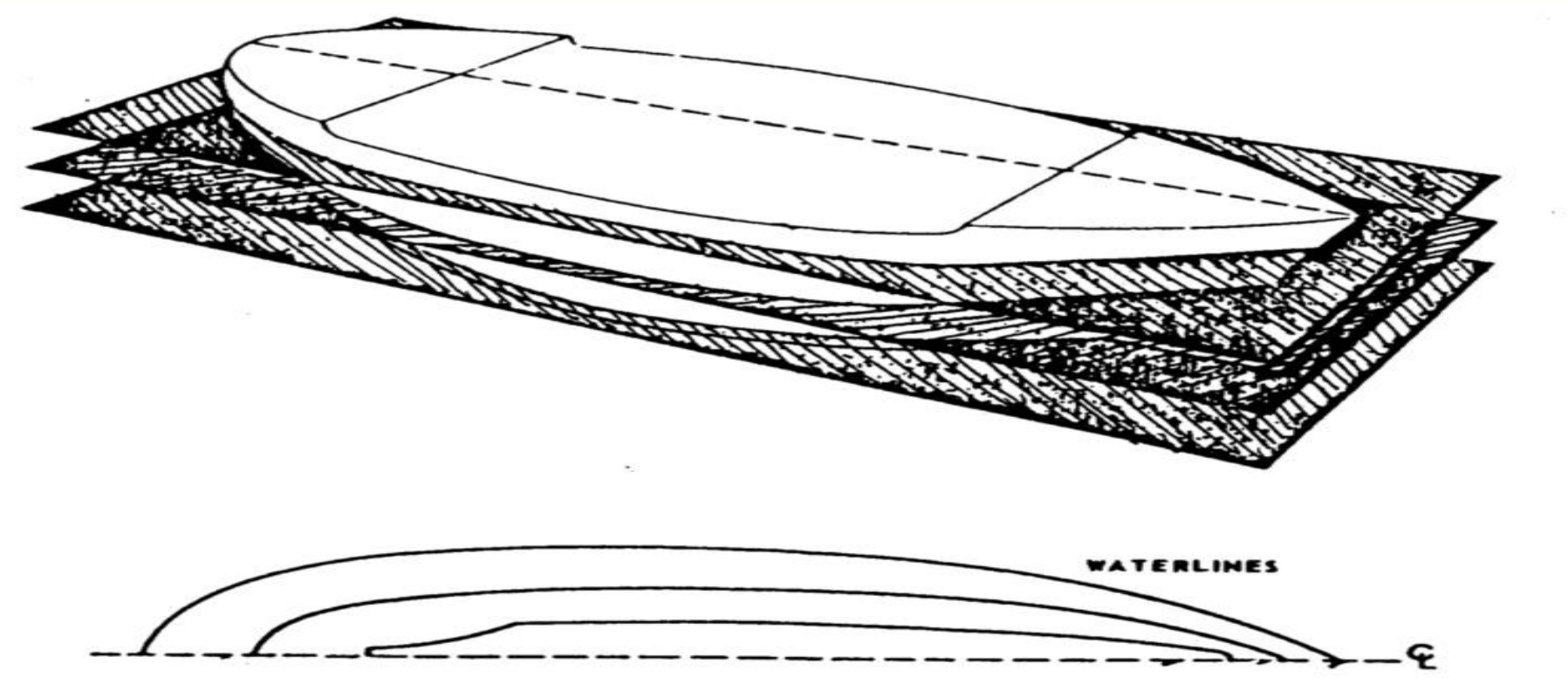
- Lines drawings/plan
  - Body plan (front View)
  - Sheer plan (side view)
  - Half breadth plan (top view)

Lines Plan

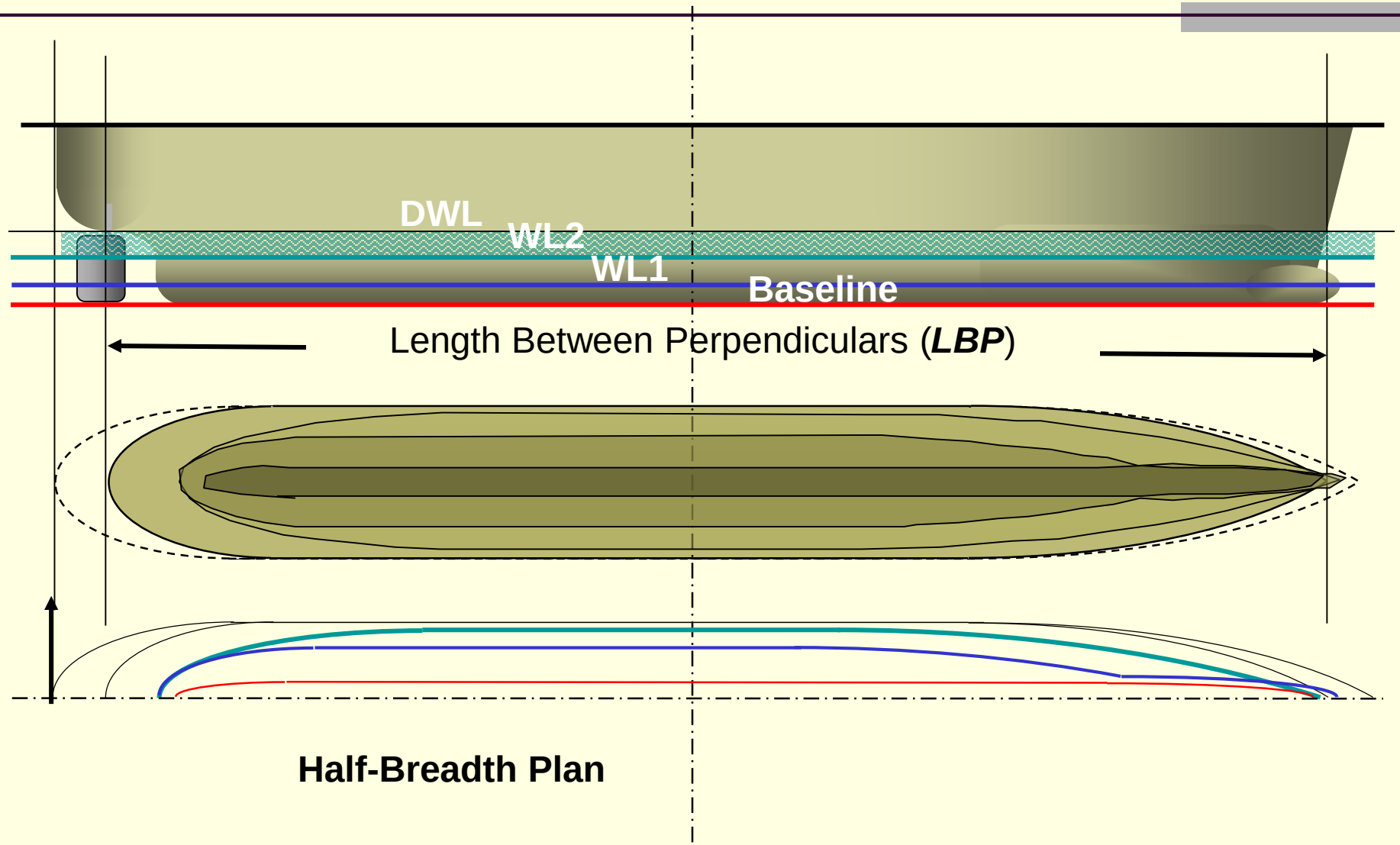

HALF-BREADTH PLAN
WATERLINES
SECTION LINES
BODY PLAN
FWD
AFT
BUTTOCK LINES
SHEER PLAN

Half – Breadth Plan

- Intersection of planes (waterlines) parallel to the baseline (keel).

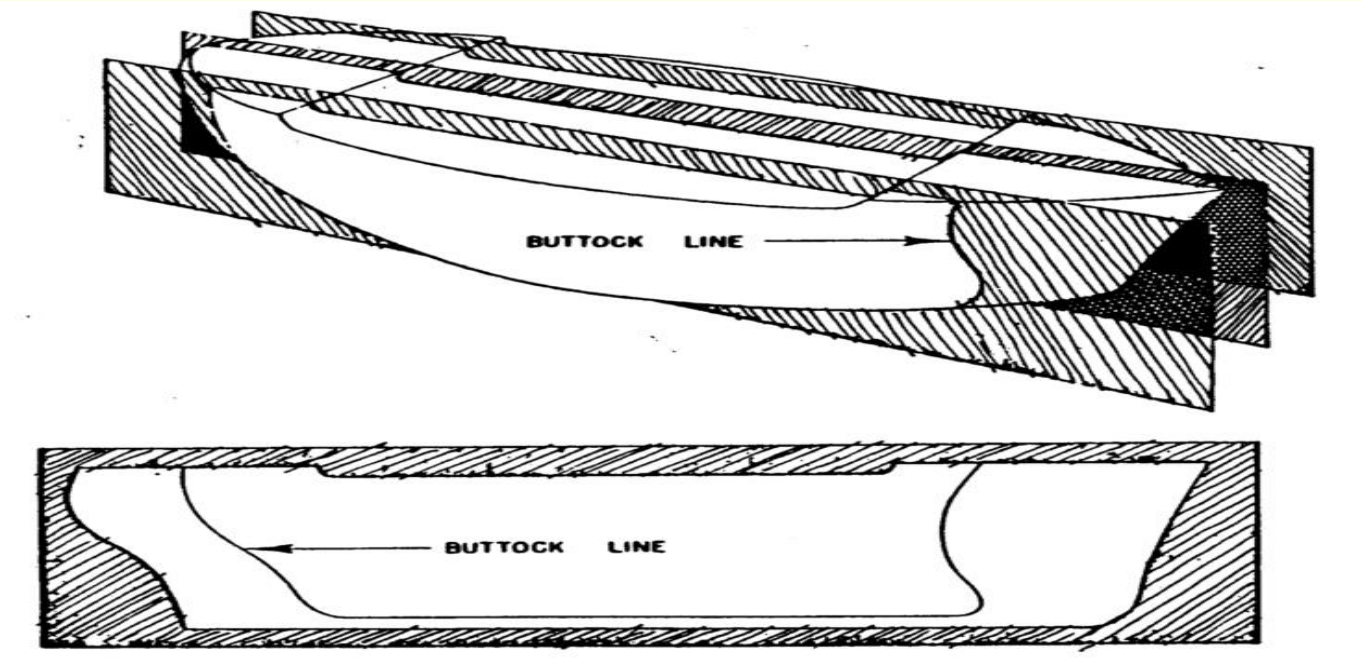


Lines Drawings: *waterlines*

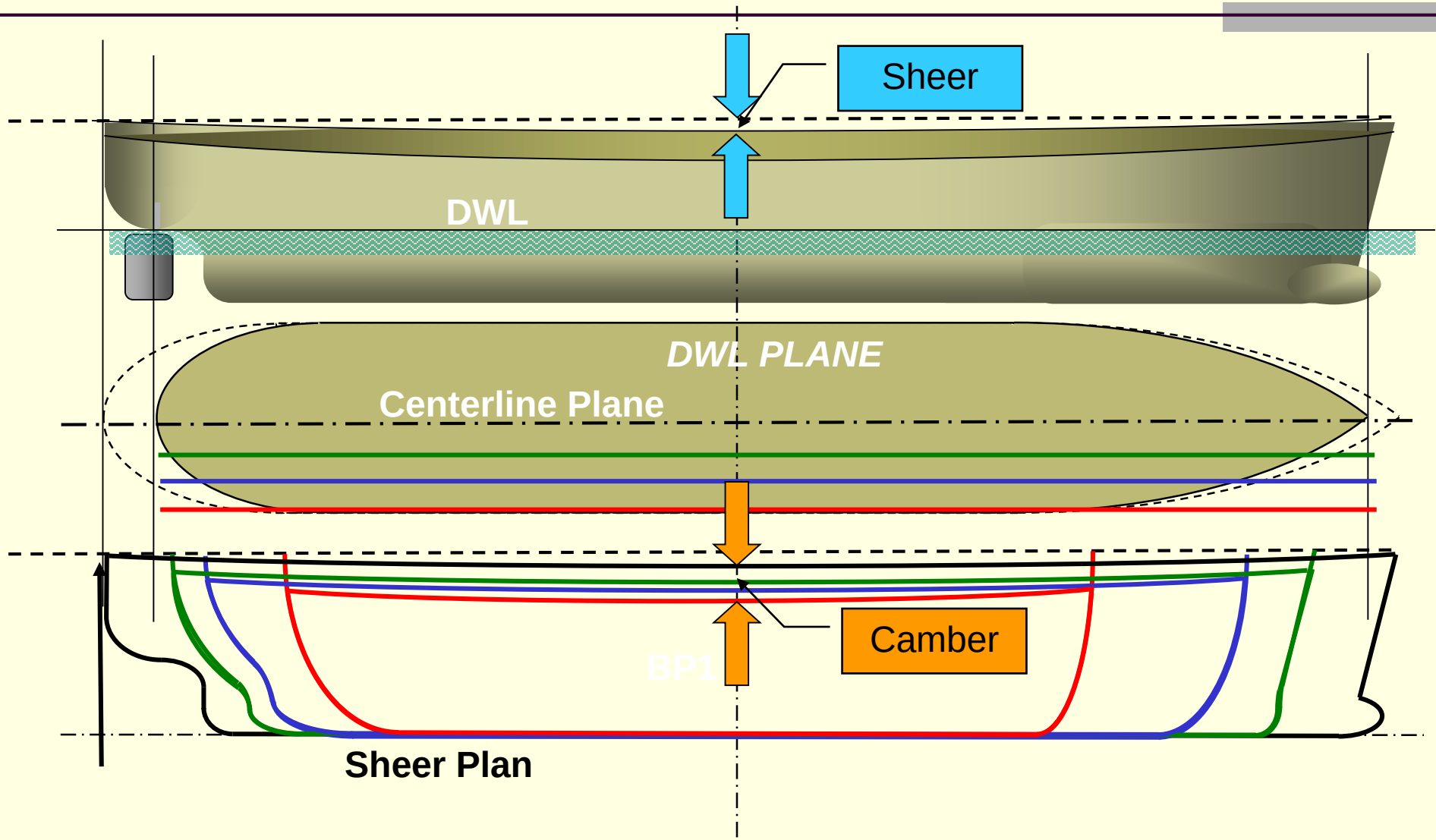


Sheer Plan

- Intersection of planes (buttock lines) parallel to the centreline plan.

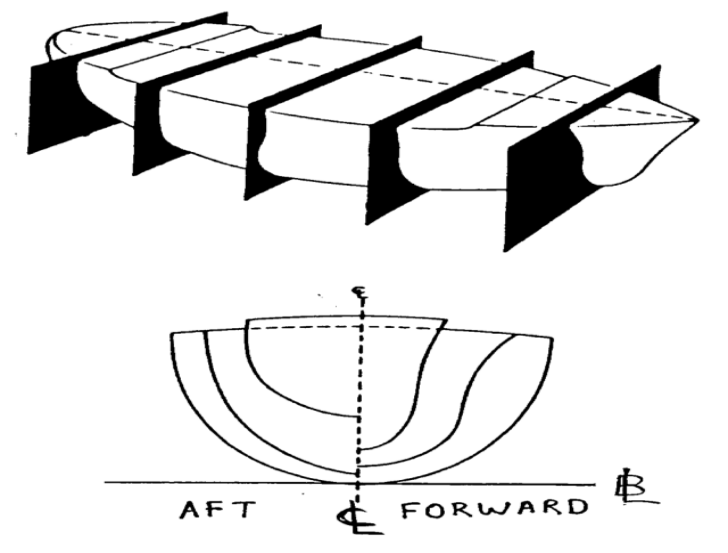


Sheer & Camber

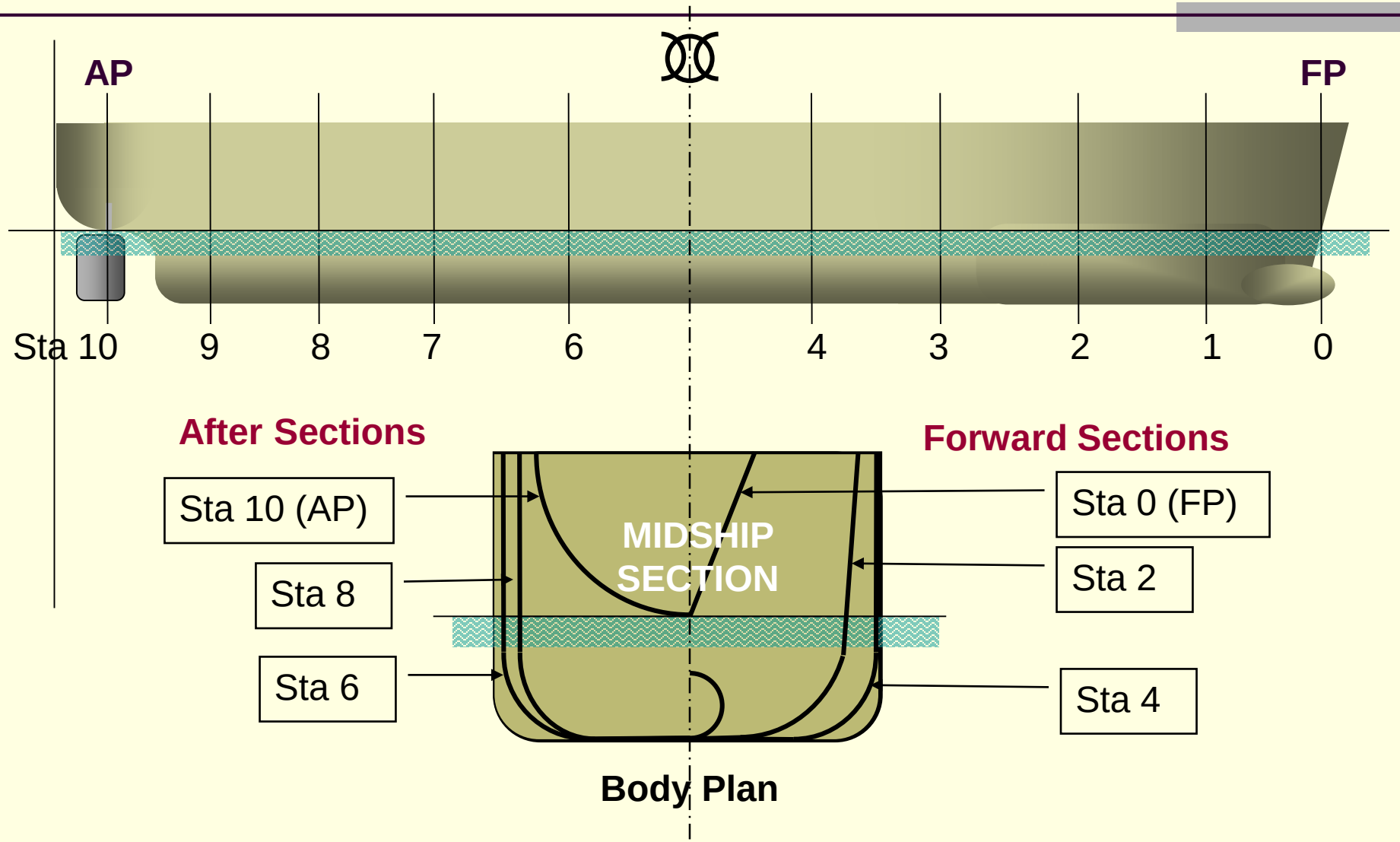


Body Plan

- Intersection of planes to define section line.
- Sectional lines show the true shape of the hull form.
- Forward sections from amidships : R.H.S.
- Aft sections from amidship : L.H.S.

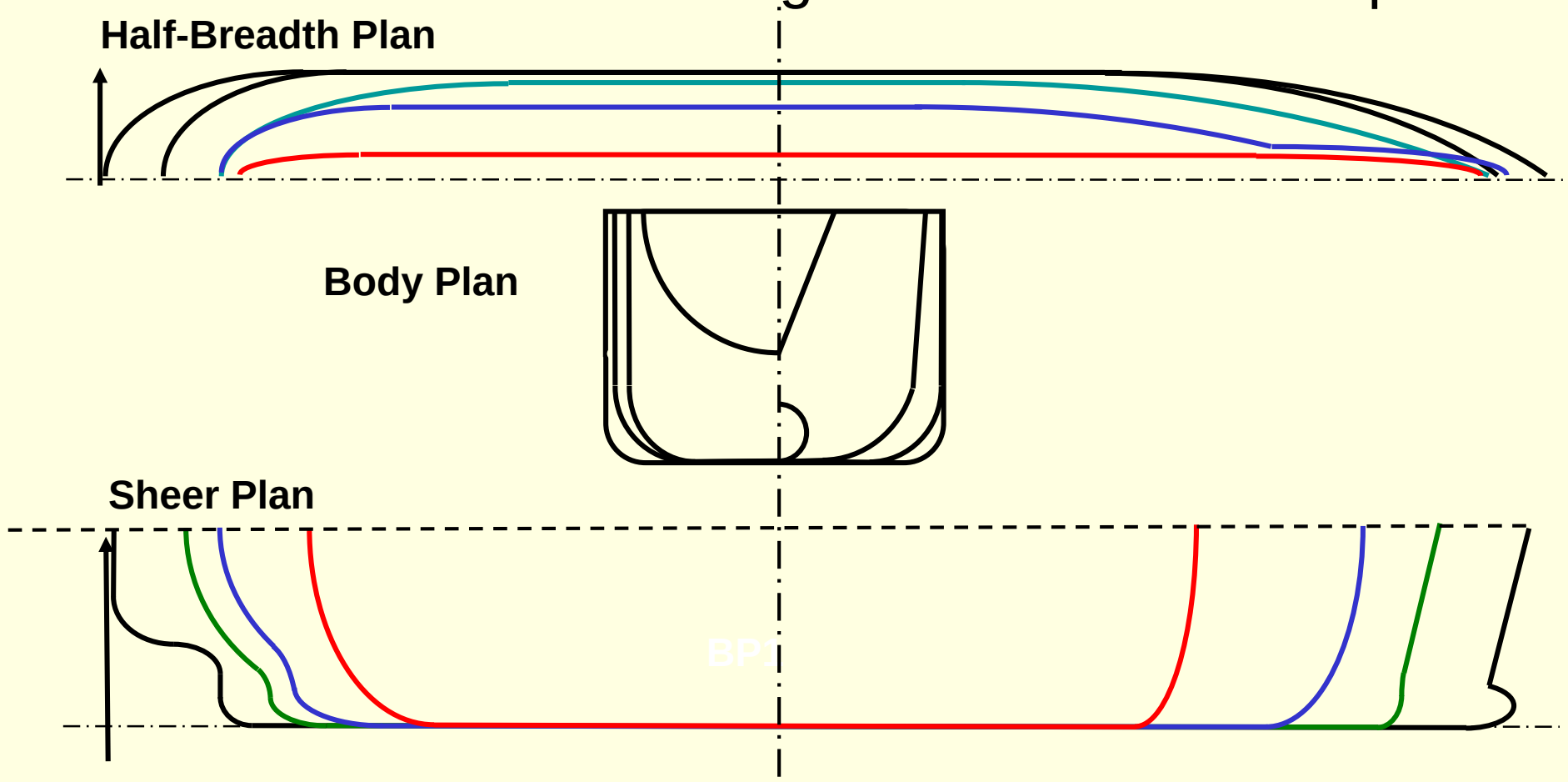


Lines Drawings: *sections*



Lines Drawings

- Three sets of drawings define the hull shape



Any Questions, Suggestions, Comments, etc.

THE END

